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EDUCATION

- **University of Pittsburgh (Swanson School of Engineering/Frederick Honors Col.)** **Pittsburgh, PA**
B.S. - Computer Engineering (Autonomous Systems Focus); GPA: 3.99/4.00 *August 2022 - April 2026*
- **Spring-Ford Area High School** **Royersford, PA**
High School Degree (Summa Cum Laude); GPA: 100.88/100.00 *August 2018 - June 2022*

PROFESSIONAL EXPERIENCE

- **ESTAT Actuation** **Pittsburgh, PA**
Robotics Developer Co-op *May 2025 - Present*
 - Building and commissioning multiple robotic test stands expanding clutch testing capacity by 133%, covering mechanical assembly, electrical wiring, programming, and documentation for long-term use and potential sales to external customers.
 - Directing the assembly and delivery of 6 custom battery boxes for the Suzuki Moqba robot at the Japan Mobility Show. Overseeing quality control, leading debugging efforts to resolve heating issues, and ensuring a professional final build.
 - Testing new generation rotary clutch prototypes to improve reliability and evaluate longevity. Creating documentation explaining testing procedures, and using these prototypes to inform design decisions for next generation production clutches.
- **Carnegie Mellon University Robotics Institute** **Pittsburgh, PA**
Undergraduate Roboticist Researcher *April 2023 - Present*
 - **ZOË 2 Rover (Advisor: Dr. David Wettergreen):** (tinyurl.com/zoe2rover)
 - * Assisting in the development of software architecture and the robotic controller implementation for a novel 4-wheeled rover with a unique passive steering mechanism.
 - * Enabling low-level software development for the rover using ROS2 and C++ to interface with motor controllers and encoders via CANOpen protocol.
 - **Underwater Snake Robot (Advisor: Dr. Howie Choset):** (tinyurl.com/humrsCMU)
 - * Implemented High-Frequency Injection methods in Brushless DC thrusters to achieve control at low/zero speeds thus reducing minimum speed by 80% allowing for reduced oscillations in station-keeping control loops.
 - * Deployed station-keeping feature using AprilTags, IMU readings, and Nested PID Controllers to perform robot state-estimation underwater.
 - * Conducted major repairs on the buoyancy module of the robot and assisted in continual electrical/hardware maintenance.
 - **Apple's E-Waste Recycling Project (Advisor: Dr. Matthew Travers):** (tinyurl.com/applecmu)
 - * Labeled datasets consisting of 4000+ images for Machine Learning Models to detect screws in e-waste images.
 - * Integrated ROS1 and Python packages to track AprilTags using a Realsense camera for localization of UR3 robotic arm.
 - * Manufactured custom AprilTags using lasercutters and sheet metal manufacturing methods.

LEADERSHIP & ACTIVITIES

- **Robotics and Automation Society (University Rover Challenge)** **Pittsburgh, PA**
Chief Engineer/Integration Lead *August 2022 - Present*
 - Leading a multidisciplinary team of 30 students in engineering a rover, coordinating integration efforts across mechanical, electrical, software, and bioscience teams. (tinyurl.com/roverimages)
 - Mentoring younger students in Onshape, KiCAD, and general manufacturing processes through hands-on building.
 - Securing and managing over \$7,000 in funding to support team development and future growth.
 - Proposed and spearheaded the development of a prototype robotic hand utilizing pneumatics. (tinyurl.com/hydraarm)
 - Securing and managing over \$7,000 in funding to support team development and future growth.
- **Formula SAE - Panther Racing** **Pittsburgh, PA**
R&D Engineer *August 2022 - March 2024*
 - hello
- **FIRST Robotics** **Exton, PA**
Team Captain/Lead Mechanical Design Engineer *January 2020 - August 2022*
 - Led a team of 40 students and qualified for the Worlds level of competition, the highest win percentage since 2005, and a top 5% ranking globally. (FIRST Link: <https://tinyurl.com/dewbot17>)
 - Utilized Solidworks to design and develop competition and award-winning robots.
- **VEX Robotics** **Royersford, PA**
Team Captain *August 2018 - June 2022*
 - Organized VEX robotics competition event for 60+ teams, hosted workshops to teach CAD to fellow club members, and received recognition for outstanding leadership. (VEX Link: <https://bit.ly/3w7a6b1>)
 - Qualified for the State's level of competition for all 4 years of high school.

PROJECTS

- **555 Timer Roulette Board (KiCAD, Onshape, Circuit Design):** Designed and built a PCB-based LED Roulette Wheel using a 555 timer and CD4017 counter, featuring dynamic speed control through capacitive timing, dip switch power control, and a custom 3D-printed enclosure for portability. (tinyurl.com/555roulette)
- **Bop It Project (KiCAD, Onshape, Microcontrollers):** Developed an interactive “Space Invaders Bop-It” reflex game using a bare ATmega328P chip, integrating analog inputs, audio-visual feedback, and custom embedded logic for randomized commands and progressively challenging gameplay. (tinyurl.com/spacebopit)
- **Micromouse Robot:** Designed, assembled, and programmed a custom Micromouse robot through iterative PCB development, integrating motor control, IMU, IR sensors, and a display system with low-level embedded programming. Implemented PID control loops for navigation. Placed 3rd in competition at Stony Brook University IEEE conference. (tinyurl.com/goyalmm)
- **VoltVitals Embedded Board:** Designed and prototyped an ESP32 based circuit with web interface to measure home outlet voltage and report over Wi-Fi, collaborating with FirstEnergy engineers to enable early outage detection. (tinyurl.com/voltvitals)
- **An Artistic take on Bipedal Robots:** Created a bipedal robot using Onshape and Arduino to carry a symbolic lantern given to all freshmen, reimagining a campus tradition through an artistic and technological lens. (tinyurl.com/gbipedal)
- **STM-32 Elevator Simulator (ARM Assembly, Project Integration):** Designed and Implemented software architecture in ARM Assembly to operate a physical elevator simulator PCB. (tinyurl.com/stmelevator)
- **Custom Cane (Human Centered Design, Solidworks):** Designed and fabricated a Walker-Cane Fusion to increase bathrooms accessibility for wheelchair users. Won first place at the Senior Design Expo. (tinyurl.com/CustomCane)
- **Formula SAE E-Brake Bias:** Developed a 3D-printed e-brake bias system for a Formula SAE racecar using Solidworks, improving performance and usability; won the FSAE Innovation Award for implementation of the project. (tinyurl.com/ebrakeb)
- **Silicon Prosthetic Hand (High School Independent Research, Solidworks):** Designed and manufactured prototype prosthetic hand with silicone soft actuators and tested with human participants for an AP Research Project. (Link: <https://tinyurl.com/myprosthetic>)

PUBLICATIONS

- M. Nye, A. Raji, A. Saba, E. Erlich, R. Exley, **A. Goyal**, A. Matros, R. Misra, M. Sivaprakasam, M. Bertogna, D. Ramanan, and S. Scherer, “**BETTY Dataset: A Multi-modal Dataset for Full-Stack Autonomy**,” in Proc. IEEE Int. Conf. Robot. Autom. (ICRA), 2025. [Online]. Available: (arxiv.org/abs/2505.07266)

SKILLS AND AWARDS

- **Software:** Linux, ROS1, ROS2, Github, Python, C++, C, MATLAB, ARM and RISC-V Assembly, CANOpen, Google Colab
- **Electrical:** KiCAD, LTSpice, Soldering, Arduino, ESP32
- **Mechanical:** Solidworks, Onshape, Milling, Lathe, MIG Welding, Laser Cutting, General Shop Tools
- **Awards:** Micromouse Competition 3rd place, SSOE Design Expo 1st place Art of Making, FSAE Innovation Award, Eagle Scout